[BUG] `qml.ctrl` broken differentiability with parameter-shift finite-diff

https://github.com/PennyLaneAI/pennylane/issues/2237

ROW 98

[BUG] `qml.ctrl` broken differentiability with parameter-shift/finite-diff #2237

New issue

Closed #2238

antalszava opened on Feb 26, 2022 · edited by antalszava

Edits ···

Expected behavior

`qml.ctrl` supports differentiating over input parameters to the function to be controlled.

For example, for an operation being controlled, the following works:

```
# Autograd
# ----
import pennylane as qml
from pennylane import numpy as np

dev = qml.device("default.qubit", wires=2)

@qml.qnode(dev, diff_method='parameter-shift')
def circuit(b):
    qml.QubitStateVector(np.array([1.0, -1.0], requires_grad=False) / np.sqrt(2), wires=0)
    qml.ctrl(qml.RY, control=0)(b, wires=[1])
    #qml.CRY(b, wires=[0, 1])
    return qml.expval(qml.PauliX(0))

b = np.array(0.123, requires_grad=True)

qml.grad(circuit)(b)
```

Actual behavior

- With Autograd: error is raised
- With TF/Torch: the gradient is 0

Additional information

*Swapping out to a dedicated controlled operation (e.g., `CRY`) solves the issue in specific cases.

- Backprop and adjoint work as expected.

Source code

```
# TF
# ------
import pennylane as qml
from pennylane import numpy as np
import tensorflow as tf

dev = qml.device("default.qubit", wires=2)

@qml.qnode(dev, interface='tf', diff_method='parameter-shift')
def circuit(x):
    qml.QubitStateVector(np.array([1.0, -1.0], requires_grad=False) / np.sqrt(2), wires=0)
    qml.ctrl(qml.RY, control=0)(x, wires=[1])
    #qml.CRY(b, wires=[0, 1])
    return qml.expval(qml.PauliX(0))

b = tf.Variable(0.123, dtype=tf.float64)
with tf.GradientTape() as tape:
    res = circuit(b)

grad1 = tape.gradient(res, b)
grad1
```

Tracebacks

```
~/xanadu/pennylane/pennylane/devices/default_qubit.py in apply(self, operations, rotations, **kwargs)
    215                 self._apply_basis_state(operation.parameters[0], operation.wires)
    216             else:
--> 217                 self._state = self._apply_operation(self._state, operation)
    218
    219         # store the pre-rotated state
```

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Fix bug to allow `qml.ctrl` to work with gradient transforms
PennyLaneAI/pennylane

Notifications Customize

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Participants

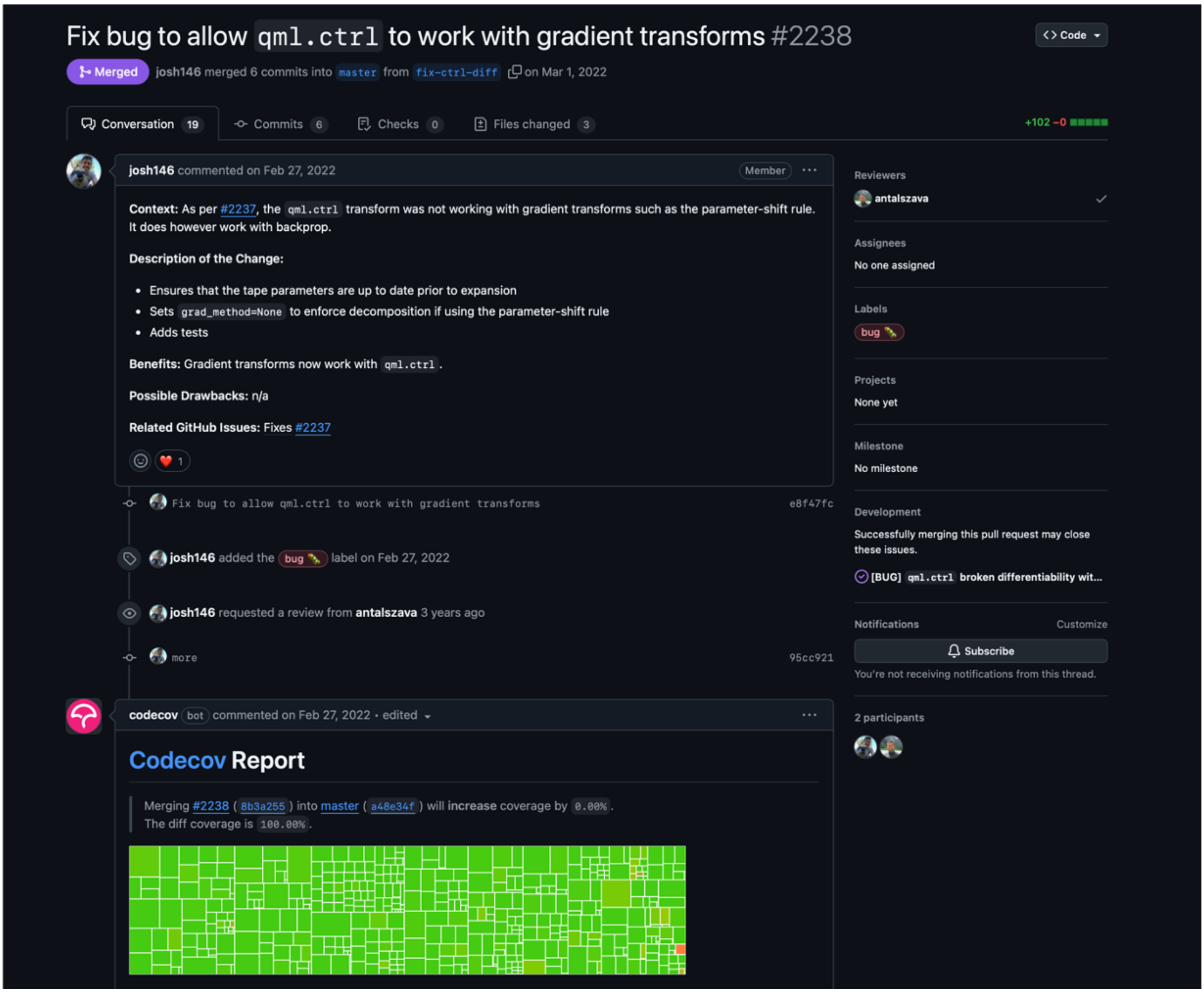
Goodnotes

```
~/xanadu/pennylane/pennylane/devices/default_qubit.py in _apply_operation(self, state, operation)
    246         if len(wires) <= 2:
    247             # Einsum is faster for small gates
--> 248             return self._apply_unitary_einsum(state, matrix, wires)
    249
    250         return self._apply_unitary(state, matrix, wires)

~/xanadu/pennylane/pennylane/devices/default_qubit.py in _apply_unitary_einsum(self, state, mat, wires)
    729         device_wires = self.map_wires(wires)
    730
--> 731         mat = self._cast(self._reshape(mat, [2] * len(device_wires) * 2), dtype=self.C_DTYPE)
    732
    733         # Tensor indices of the quantum state

TypeError: must be real number, not ArrayBox
```

System information

```
Name: PennyLane
Version: 0.22.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /xanadu/pennylane
Requires: numpy, scipy, networkx, retworkx, autograd, toml, appdirs, semantic_version, autoray, cachetools, pe
Required-by: PennyLane-Lightning, amazon-braket-pennylane-plugin, PennyLane-Orquestra, pennylane-qulacs, Penny
Platform info:           Linux-5.13.0-30-generic-x86_64-with-glibc2.10
Python version:          3.8.5
Numpy version:           1.22.1
Scipy version:           1.7.3
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.21.0)
- braket.aws.qubit (amazon-braket-pennylane-plugin-1.5.3)
- braket.local.qubit (amazon-braket-pennylane-plugin-1.5.3)
- orquestra.forest (PennyLane-Orquestra-0.15.0)
- orquestra.ibmq (PennyLane-Orquestra-0.15.0)
- orquestra.qiskit (PennyLane-Orquestra-0.15.0)
- orquestra.qulacs (PennyLane-Orquestra-0.15.0)
- qulacs.simulator (pennylane-qulacs-0.17.0.dev0)
- honeywell.hqs (PennyLane-Honeywell-0.16.0.dev0)
- aqt.noisy_sim (PennyLane-AQT-0.18.0)
- aqt.sim (PennyLane-AQT-0.18.0)
- projectq.classical (PennyLane-PQ-0.18.0.dev0)
- projectq.ibm (PennyLane-PQ-0.18.0.dev0)
- projectq.simulator (PennyLane-PQ-0.18.0.dev0)
- forest.numpy_wavefunction (PennyLane-Forest-0.18.0.dev0)
- forest.qvm (PennyLane-Forest-0.18.0.dev0)
- forest.wavefunction (PennyLane-Forest-0.18.0.dev0)
- microsoft.QuantumSimulator (PennyLane-qsharp-0.19.0)
- ionq.qpu (PennyLane-IonQ-0.17.0.dev0)
- ionq.simulator (PennyLane-IonQ-0.17.0.dev0)
- strawberryfields.fock (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gaussian (PennyLane-SF-0.20.0.dev0)
- strawberryfields.gbs (PennyLane-SF-0.20.0.dev0)
- strawberryfields.remote (PennyLane-SF-0.20.0.dev0)
- strawberryfields.tf (PennyLane-SF-0.20.0.dev0)
- cirq.mixedsimulator (PennyLane-Cirq-0.20.0.dev0)
- cirq.pasqal (PennyLane-Cirq-0.20.0.dev0)
- cirq.qsim (PennyLane-Cirq-0.20.0.dev0)
- cirq.qsimh (PennyLane-Cirq-0.20.0.dev0)
- cirq.simulator (PennyLane-Cirq-0.20.0.dev0)
- qiskit.aer (PennyLane-qiskit-0.22.0.dev0)
- qiskit.basicaer (PennyLane-qiskit-0.22.0.dev0)
- qiskit.ibmq (PennyLane-qiskit-0.22.0.dev0)
- qiskit.ibmq.circuit_runner (PennyLane-qiskit-0.22.0.dev0)
- qiskit.ibmq.sampler (PennyLane-qiskit-0.22.0.dev0)
- default.gaussian (PennyLane-0.22.0.dev0)
- default.mixed (PennyLane-0.22.0.dev0)
- default.qubit (PennyLane-0.22.0.dev0)
- default.qubit.autograd (PennyLane-0.22.0.dev0)
- default.qubit.jax (PennyLane-0.22.0.dev0)
- default.qubit.tf (PennyLane-0.22.0.dev0)
- default.qubit.torch (PennyLane-0.22.0.dev0)
```

Existing GitHub issues



Fix bug to allow `qml.ctrl` to work with gradient transforms #2238

Code

Merged josh146 merged 6 commits into `master` from `fix-ctrl-diff` on Mar 1, 2022

Conversation 19 · Commits 6 · Checks 0 · Files changed 3 · +102 −0

josh146 commented on Feb 27, 2022 · Member

Context: As per #2237, the `qml.ctrl` transform was not working with gradient transforms such as the parameter-shift rule. It does however work with backprop.

Description of the Change:

- Ensures that the tape parameters are up to date prior to expansion
- Sets `grad_method=None` to enforce decomposition if using the parameter-shift rule
- Adds tests

Benefits: Gradient transforms now work with `qml.ctrl`.

Possible Drawbacks: n/a

Related GitHub Issues: Fixes #2237

❤️ 1

`Fix bug to allow qml.ctrl to work with gradient transforms` e8f47fc

josh146 added the bug label on Feb 27, 2022

josh146 requested a review from **antalszava** 3 years ago

`more` 95cc921

codecov bot commented on Feb 27, 2022 • edited

Codecov Report

> Merging #2238 (`8b3a255`) into master (`a48e34f`) will **increase** coverage by `0.00%`.
> The diff coverage is `100.00%`.

Reviewers: antalszava

Assignees: No one assigned

Labels: bug

Projects: None yet

Milestone: No milestone

Development: Successfully merging this pull request may close these issues.

[BUG] `qml.ctrl` broken differentiability wit...

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2 participants